

Traceable AI-Assisted RAN Slicing: Mobility-Selective Soft Slicing in 6G RAN Using Proximal Policy Optimization

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Abstract—6G networks must allocate shared spectrum across heterogeneous services (mIoT/eMTC, URLLC, eMBB). Hard slicing provides isolation but sacrifices efficiency [?]; in our experiments, 5–8% loss (81–84% hard vs. 87% aggregated, 79.5–84.3% hard vs. 82.1–91.2% soft PPO). Aggregation violates fairness and SLA. Soft slicing with dynamic borrowing offers middle ground but requires algorithmic optimization. Novel insight: mIoT/eMTC are stationary while eMBB/URLLC are mobile. Prior RL work treats mobility generically, missing this asymmetry. We formulate spectrum allocation as an MDP and solve via PPO with mobility-selective state design: separate $v_{\text{eMBB}}, v_{\text{URLLC}}$, with v_{eMTC} omitted. This enables the learned policy to anchor on stable mIoT/eMTC floors, freeing spectrum for dynamic eMBB/URLLC borrowing. Small neural network (64→64→336, $< 1 \mu\text{s}$ inference) trained via four-phase curriculum learning. Results across diverse scenarios: 84.7% soft PPO efficiency (vs. 81.2% hard), 99.99999% URLLC SLA maintained, +5.2% revenue uplift, 20–30% lower allocation churn, 25% faster convergence (5,400 episodes). RAN neutrality preserved via transparent tariffing.

Keywords: RAN Slicing, Soft Partitioning, PPO, Mobility-Aware Allocation, 6G, Heterogeneous Services.

I. INTRODUCTION

Heterogeneous 6G networks require partitioning shared spectrum to serve mIoT (low bandwidth, massive scale), URLLC (sub-1 ms latency, 99.99999% reliability), and eMBB (elastic throughput). Hard slicing guarantees isolation but sacrifices efficiency; pure aggregation maximizes efficiency but violates service fairness and exposes URLLC to congestion.

Prior work gaps: (1) Reinforcement learning for RAN slicing (Ye et al. [1], Wang et al. [2]) uses generic mobility models, ignoring that mIoT/eMTC are stationary while eMBB/URLLC are mobile; (2) soft slicing concepts (Bega et al. [3]) are proposed but lack algorithmic solutions; (3) multi-objective optimization balancing efficiency, SLA, neutrality, and profitability simultaneously is unaddressed.

Our contribution: We exploit mobility asymmetry via mobility-selective state design and Proximal Policy Optimization (PPO). By modeling mobility separately for eMBB/URLLC while omitting it for stationary eMTC, we enable the learned policy to recognize eMTC demand as predictable, allocating a stable soft floor that anchors dynamic borrowing between mobile services. Results demonstrate 84.7% efficiency (+3.5 pp over hard slicing), 99.99999% URLLC SLA, +5.2% revenue, and 25% faster convergence than generic mobility models.

II. SYSTEM MODEL & PPO FORMULATION

A. Mobility-Selective State Space

State at slot t :

$$s_t = [B_{\text{total}}, \rho_{\text{eMBB}}, \rho_{\text{URLLC}}, \rho_{\text{eMTC}}, q_{\text{eMBB}}, q_{\text{URLLC}}, q_{\text{eMTC}}, \text{SNR}, v_{\text{eMBB}}, v_{\text{URLLC}}] \quad (1)$$

Key innovation: Separate $v_{\text{eMBB}}, v_{\text{URLLC}}$ (km/h) from generic mobility. Omit v_{eMTC} (stationary). This ~ 13 -dim state guides PPO to learn that eMTC is stable—enabling aggressive dynamic borrowing.

B. Action Space & Reward

Action: $a_t = [B_{\text{URLLC}}, B_{\text{eMTC}}, B_{\text{eMBB}}]$ s.t. $\sum B_i \leq B_{\text{total}}$.

Multi-objective reward:

$$R_t = 0.40R_{\text{SLA}} + 0.25R_{\text{eff}} + 0.20R_{\text{neut}} + 0.15R_{\text{prof}} \quad (2)$$

where $R_{\text{SLA}} = +2.0$ if URLLC meets SLA, -10.0 if violated; $R_{\text{eff}} = \frac{B_{\text{used}}}{B_{\text{total}}} \times 0.5$; $R_{\text{neut}} = -0.5$ if starved; $R_{\text{prof}} = -\frac{\text{Rev}_{\text{baseline}} - \text{Rev}_{\text{sliced}}}{\text{Rev}_{\text{baseline}}} \times 100$.

C. PPO Algorithm

Policy network (actor): $\pi(a|s; \theta) = \text{softmax}(\text{NN}_{\theta}(s))$, where NN is 64→ReLU→64→ReLU→336.

Value network (critic): $V(s; \phi)$ outputs scalar value estimate.

PPO objective:

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t \left[\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right] \quad (3)$$

where $r_t(\theta) = \frac{\pi(a_t|s_t; \theta)}{\pi(a_t|s_t; \theta_{\text{old}})}$, $\epsilon = 0.2$, $\hat{A}_t = \text{GAE estimate}$ ($\gamma = 0.99, \lambda = 0.95$).

Training: Curriculum learning across 4 phases (1,350 episodes each):

- **Phase 1:** SLA foundation (abundant spectrum, $\lambda_{\text{SLA}} = 0.50$)
- **Phase 2:** Device density (eMTC-heavy, $\lambda_{\text{SLA}} = 0.45, \lambda_{\text{eff}} = 0.25$)
- **Phase 3:** Spectrum constraint (50 RBs, $\lambda_{\text{eff}} = 0.30$)
- **Phase 4:** Profitability (mixed scenarios, $\lambda_{\text{prof}} = 0.15$)

Hyperparameters: lr = 3×10^{-4} , batch=512, epochs=10, rollout=2048.

Scenario	Hard Slicing	Soft PPO	Gain
Balanced (70/15/15)	79.5%	82.1%	+2.6 pp
IoT-Heavy (60/30/10)	84.3%	91.2%	+6.9 pp
High-Mobility	~80%	85%	+5 pp
Avg. Efficiency	81.2%	84.7%	+3.5 pp

TABLE I
SPECTRUM EFFICIENCY ACROSS SCENARIOS.

III. EXPERIMENTAL RESULTS

A. Scenario Performance

URLLC SLA: 99.99999% (maintained across all scenarios).

Revenue: +5.2% uplift (IoT-heavy scenario via eMTC volume + eMBB retention).

Allocation churn: 22% of slots (vs. 30% generic mobility) — 20–30% reduction.

B. Convergence

Training on RTX 3080: **5,400 episodes** (~1.2 hrs), 25% faster than generic v_{mobile} models (7,200 episodes). Efficiency plateaus at 84.7% by episode 4,500.

C. Learned Policy (Qualitative)

PPO discovered:

- **eMTC anchor:** 8–15 RBs (stable, no fading buffer needed)
- **URLLC adaptive:** 12–20 RBs (varies with v_{URLLC} , SNR)
- **eMBB flexible:** remainder (borrows from stable eMTC)

Key insight: Recognizing eMTC stationarity frees ~2–3 RBs/slot for dynamic reallocation, explaining +3.5 pp efficiency gain.

IV. REGULATORY & BUSINESS IMPLICATIONS

RAN Neutrality: Preserved via:

- **Per-slice tariffs:** URLLC \$30/Mbps (safety-critical), eMTC \$0.50/device (volume), eMBB \$0.10/Mbps (elastic).
- **Transparent rules:** Allocation policy verifiable; no content discrimination.
- **Fairness:** Per-flow URLLC, per-device eMTC allocation minima enforced.

Business case: Revenue uplift +5.2% through eMTC monetization ($\$0.50/\text{device} \times 1\text{M devices} = \500k/month) + URLLC premium ($\$30/\text{Mbps} \times 10\text{k subscribers} = \300k/month) + eMBB retention (throughput +8% $\rightarrow$ +2% ARPU).

V. DISCUSSION & CONCLUSION

Key findings: (1) Mobility-selective state design reduces convergence by 25% via simpler feature relationships; (2) soft slicing with learned borrowing outperforms both hard slicing (rigid) and aggregation (unfair); (3) small NN ($< 1 \mu\text{s}$ inference) enables edge deployment without cloud latency.

Limitations: Assumes known traffic mix ρ_i ; online prediction required for real deployment. Multi-cell MARL (competitive borrowing) is future work.

Conclusion: Mobility-selective PPO-based soft slicing achieves 84.7% efficiency, 99.99999% SLA, and +5.2% revenue while preserving RAN neutrality. Edge-deployable small NN makes this practical for 6G carriers.

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